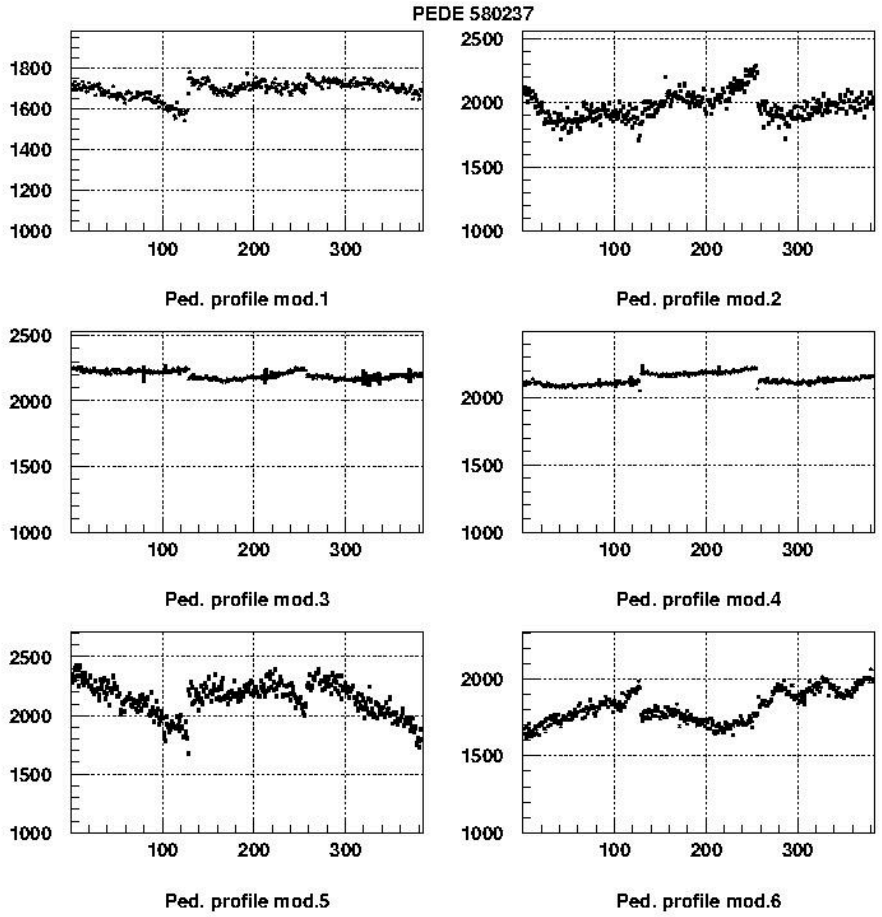
Logbook test beam CERN SELDOM H8 2023 08

2023 08 14

Test in control room of telescopes

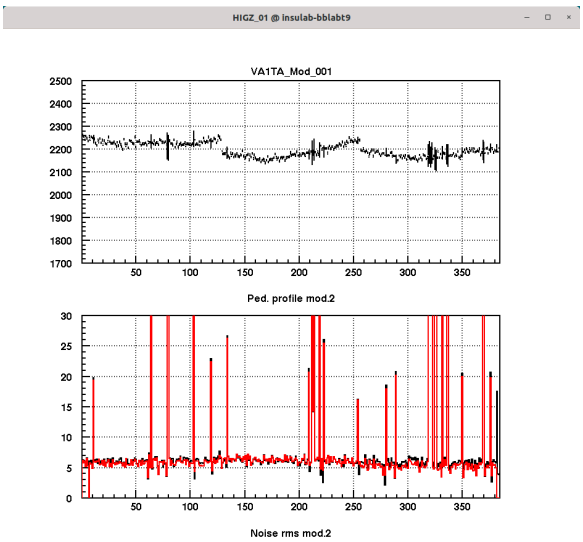
REFERENCE PLOTS

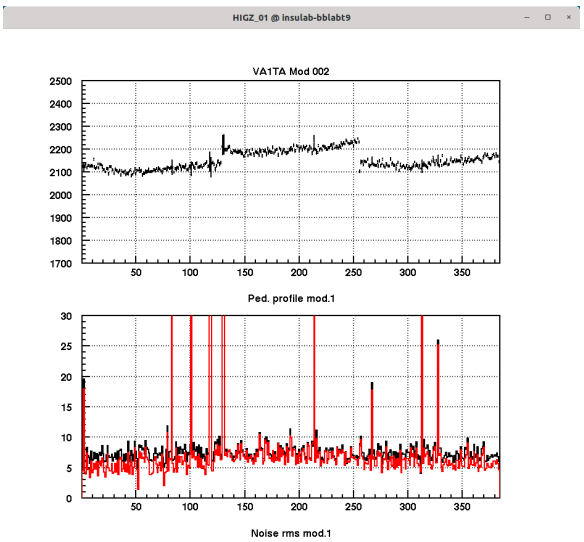


Primo plot → VA2TA
Secondo plot → 2 VA1TA
Terzo plot → modulo E

Mounting VA1TA telescope
The one with **mod 001 / mod 002**. Both channels must be crossed

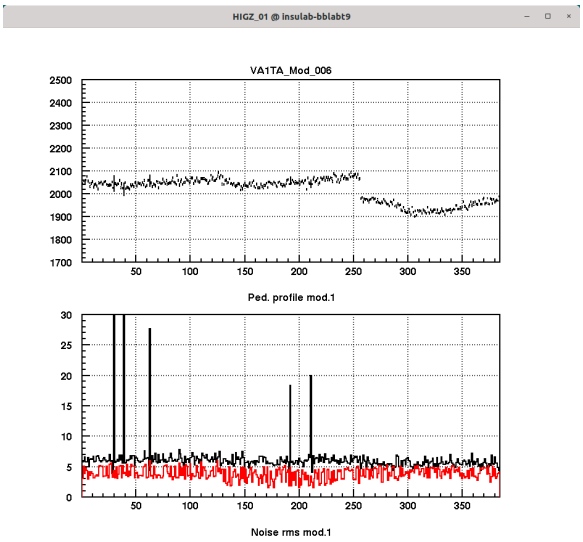
Pedestal run 401014





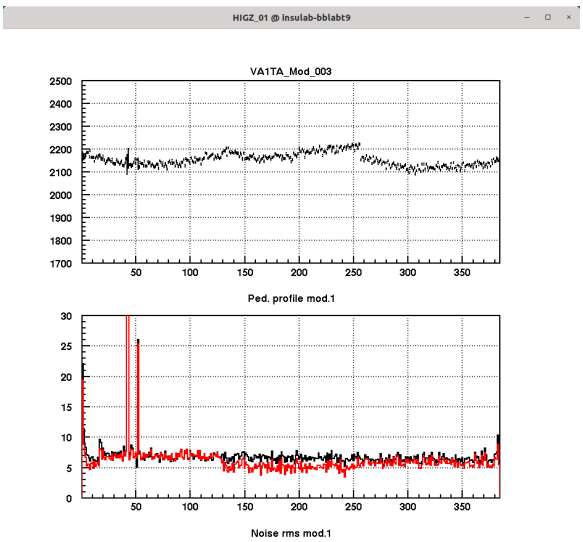
Tested trigger bits with configuration board 2202 and memory board as daughter card and everything is ok.

Mounted **mod 006**, to be used ad BACKUP
Its analog connectors are also crossed

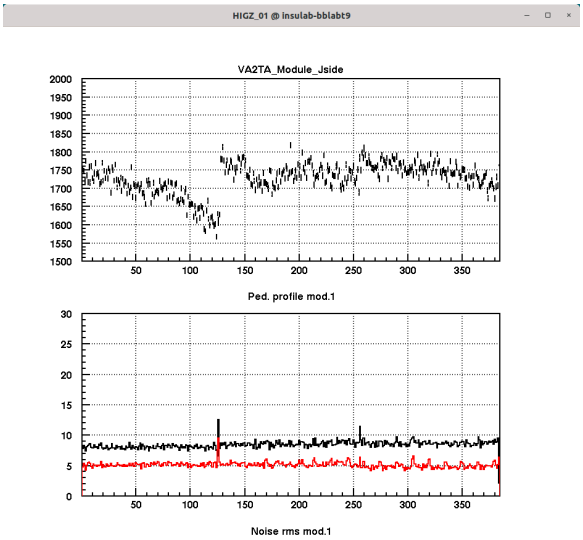


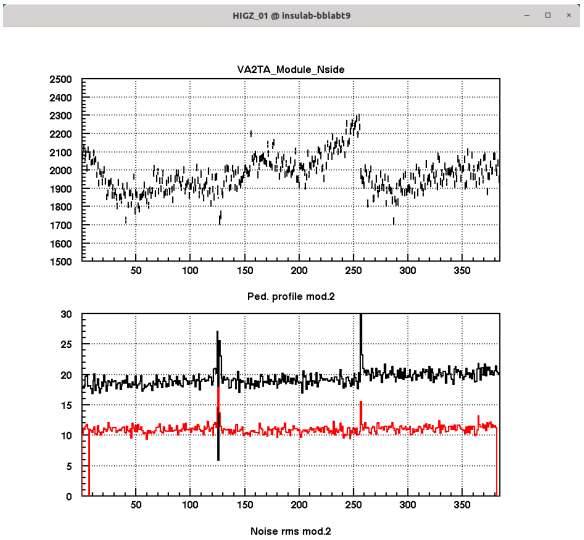
Pedestal 401015
Trigger & masks ok

Mounted **mod 003**, to be used as BACKUP
Its analog connectors are also crossed
Pedestal 401015
Trigger & masks ok



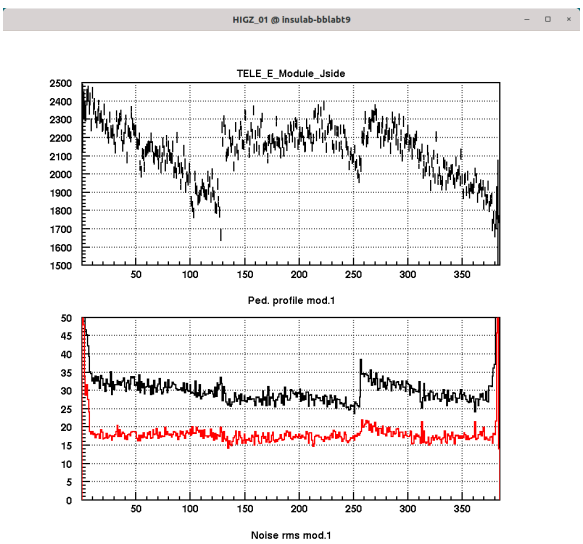
VA2TA Module
Power consumption LOW: +5: 1062mA, -5 445 mA
Power Consumption High: +5: 606 mA, -5 324 mA
Trigger and masks OK
Pedestal 401020

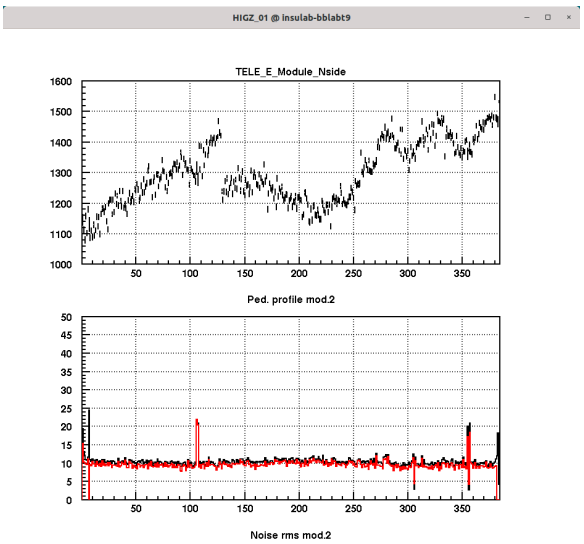




TELE E

Power consumption LOW: +5: 777mA, -5 256 mA
Power Consumption High: +5: 214 mA, -5 131 mA
Bias 37V
Pedestal 401030
Long run 401031





23/08/2023

Installation in H8 - PPE 128

Following the beam:

- VA1TA
- VA2TA
- TELE 3
- BC1
- BC2

In the readout → first va2ta and then va1ta → in the ascii files they will be in the correct order

VA1TA readout

first Y 1-top / 384-bottom
second X 1-jura / 384-saleve

VA2TA readout

first X 1-saleve / 384-jura
second Y 1-top / 384 bottom

TELE-E readout

first X 1-saleve / 384-jura
second Y 1-top / 384 bottom

BC4 readout (the farthest one)

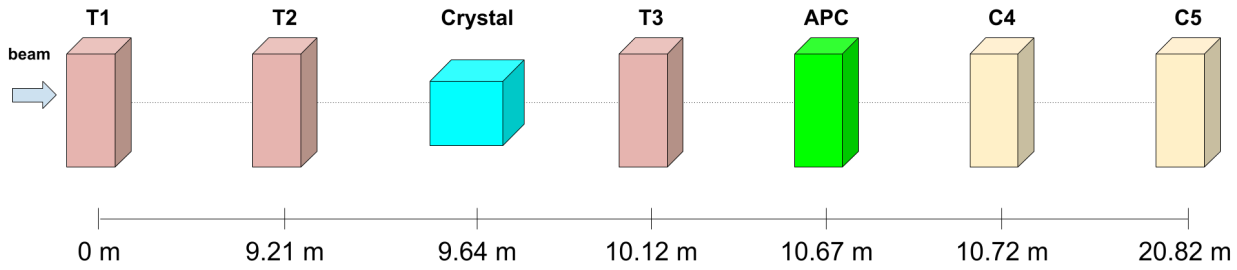
first Y 1-bottom / 384-top
second X 1-saleve / 384 jura

BC5 readout (the one used if not BC2)

second Y 1-bottom / 384-top
first X 1-saleve / 384-jura

REFERENCE SYSTEM
X from Saleve to Jura
Y from bottom to top

Experimental setup:



Recap of the most important distances:

- T1 - T2: 9.21 m
- T2 - Crystals (on goniometer): 0.43 m
- T2 - T3: 0.91 m
- T3 - C4: 0.60 m
- C4 - C5: 10.10 m

Notes:

1. All distances are measured from the center of the first object to the center of the second
2. T* indicates a silicon telescope (~ 2 cm x 2 cm active area, double-sided, ~ 7 um spatial resolution on the junction side)
3. C* indicates a silicon chamber (~ 10 cm x 10 cm active area, pair of single-sided planes, ~ 30 um spatial resolution on both sides)
4. APC is a plastic scintillator

720154 pede BC front
720155 pede BC back

Problems with the third telescope and the chamber → they are due to the power supplies → need to change them? Let's see how it works tonight

CHAMBER FAR from here on
720189 pede
720191 alignment run to be compared with 720091 → first telescope ok, second telescope to move, third no idea where it is

Tele 2 moved

- Due to inverson in VA2TA ASIC, previously moved 1cm toward Jura. Now we moved 2cm toward saleve to correct. It is perfect
- y direction. The order was to move 1 cm up. Looking ai segni sulla rotaia, sembra chee fossero scesi 1cm. We decided to move 1 cm more toward saleve. Good but not centred

720195 pede
720196 beam

New movement
- Tele2 (VA2TA): .5 cm down
- Tele3 1.5 cm Salève

720197 pede
720198 beam

Now we must do
- Tele2 (VA2TA): 1 cm up
- LATEST CHAMBER 1.5 cm Jura

And we did it like that!

720199 pede
720200 beam
720201 beam
720202 pede with 3k particles per spill

720203 pede

Tele 2:.5 cm down

720206 pede
720207 beam ALIGNED!!!!

720208 beam zero supp
720209 pede 2.5MHz
720210 pede tele3 5MHz
720211 beam

720212 Trig sel 3, should be x

Trig sel 3: y of 1st telescope (VA1TA)
Trig sel 4,

720213 Aborted, wrong trig sel
720214 Trg sel 2. Confirmed it's x

Where data are

Data saved by the DAQ are directly uploaded to the following CERNBOX path:
/eos/project/i/insulab-como/testBeam/TB_2023_08_H8_SELDOM

To be allowed to read from this directory, you should be member of
cernbox-project-insulab-como-readers
Ask Stefano (scarsi@studenti.uninsubria.it) for more info

There exists also a public link, which everyone can use to download files
https://cernbox.cern.ch/s/6MFd2Ujs2N6UM3k

Three folders are available in the path linked. In each folder, data are saved in a different format:

- ASCII : both ascii files produced by the stripping and hbook raw data
- HDF5
- ROOT TTree files

Data may be either analyzed

- Directly online (using SWAN or a virtual machine)
- Mounting a folder via sshfs
  sshfs -o ro
  yourname@lxplus.cern.ch:/eos/project/i/insulab-como/testBeam/TB_2023_08_H8_SELDOM .
  Mind the final point, representing the mounting point on your pc
- offline downloading the sources

ASCII STRUCTURE

- 8 xpos (x1, y1, x2, y2, x3, y3, x4, y4)
- Num strip hit in the cluster
- 5 info about goniometer positions [ROT, CRAD, LAT LIN, LAT BIG, NONE]
- 2 info plus (Step gonio + num spill)
- 1 Numero di evento

HDF5 / ROOT files structure
They contains the same information present in ASCII files, but with different matrices inside.
Each point of the bullet list above is a different matrix
Check pics below

```
(base) scarsi@scarsivm8:/eos/project/i/insulab-como/testBeam/TB_2023_08_H8_SELDOM/ROOT $ root -l run720214.ROOT
root [0]
Attaching file run720214.ROOT as _file0...
(TFile *) 0x369e9b0
root [1] .ls
TFile**         run720214.ROOT
 TFile*         run720214.ROOT
  KEY: TTree    t;1
root [2] t->Print()
******************************************************************************
*Tree    :t         :                                                       *
*Entries :    78292 : Total =        15090723 bytes  File  Size =    3123233 *
*        :          : Tree compression factor =   4.84                      *
******************************************************************************
*Br    0 :xpos      : xpos[8]/D                                             *
*Entries :    78292 : Total  Size=    5022091 bytes  File Size  =    2456576 *
*Baskets :      126 : Basket Size=      32000 bytes  Compression=   2.04     *
*.............................................................................*
*Br    1 :nstrip    : nstrip[8]/L                                           *
*Entries :    78292 : Total  Size=    5022351 bytes  File Size  =     419105 *
*Baskets :      126 : Basket Size=      32000 bytes  Compression=  11.98     *
*.............................................................................*
*Br    2 :xinfo     : xinfo[5]/D                                            *
*Entries :    78292 : Total  Size=    3143213 bytes  File Size  =      44346 *
*Baskets :      126 : Basket Size=      32000 bytes  Compression=  70.82     *
*.............................................................................*
*Br    3 :info_plus : info_plus[2]/L                                        *
*Entries :    78292 : Total  Size=    1264725 bytes  File Size  =      25235 *
*Baskets :      126 : Basket Size=      32000 bytes  Compression=  50.00     *
*.............................................................................*
*Br    4 :ievent    : ievent/L                                              *
*Entries :    78292 : Total  Size=     637993 bytes  File Size  =     172789 *
*Baskets :      126 : Basket Size=      32000 bytes  Compression=   3.68     *
*.............................................................................*
root [3]
```

```
# Frascati 2023 Settings
hf.create_dataset("xpos", data =  df.iloc[:,0:4], maxshape=(None,4), **opts)
hf.create_dataset("nstrip", data =  df.iloc[:,4:8], maxshape=(None,4), **opts)
hf.create_dataset("nclu", data =  df.iloc[:,8:12], maxshape=(None,4), **opts)

hf.create_dataset("digiBase", data =  df.iloc[:,12:28], maxshape=(None,16), **opts)
hf.create_dataset("digiPH", data =  df.iloc[:,28:44], maxshape=(None,16), **opts)
hf.create_dataset("digiTime", data =  df.iloc[:,44:60], maxshape=(None,16), **opts)

hf.create_dataset("xinfo", data =  df.iloc[:,60:65], maxshape=(None,5), **opts)
hf.create_dataset("info_plus", data =  df.iloc[:,65:67], maxshape=(None,2), **opts)
hf.create_dataset("ievent", data =  df.iloc[:,67], maxshape=(None,), **opts)
```

720217 Try to trigger on va2ta - Thr 160 - trig sel 1, range 192-256
720218 172-212 to center on beam
confirm that trig sel 1 is x

720219 Trig sel 8: and of ors, to create the tube

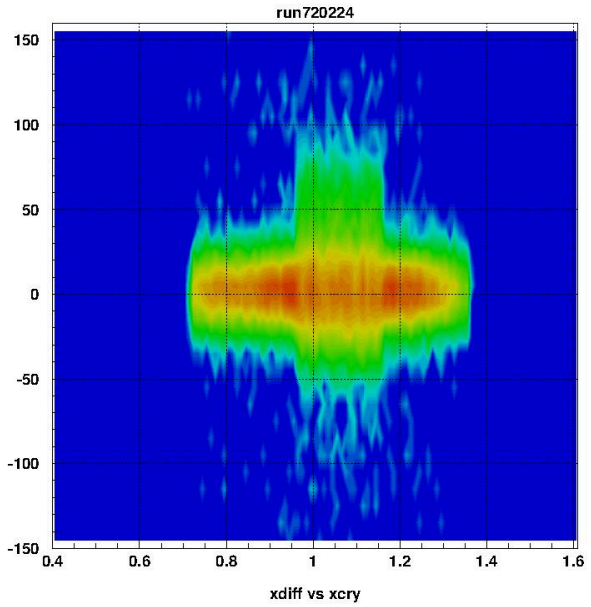
Moved BC4 and APC out of beam

CRYSTAL STF-139
(crystal for LHC beam extraction)

720222 Scan -2cm +2cm step 1mm to find the crystal.9000ev per step = 1 min per punto
see scattering starting from step 21
Apparently we see the C-holder
Try to do a finer scan starting from the holder and going negative (Jura)

720223 Andiamo a 0. Da -50 a -20, step 2 = 1mm
The crystal enters on the 5th step, which is -42 lat big
8th step is centered (-36)

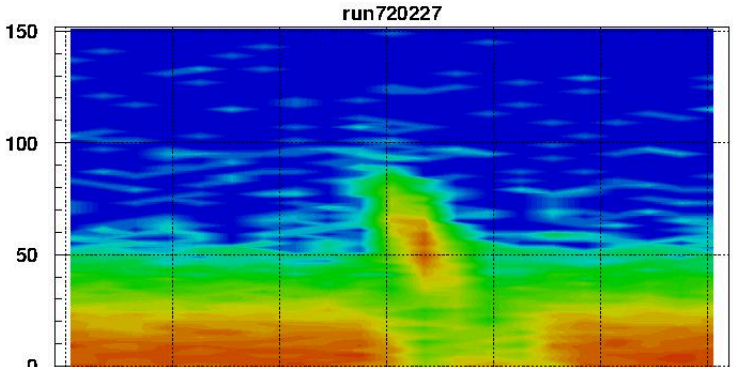
Therefore we go to -36



x axis = x position on the crystal plane
y axis = deflection

720224 Single daq, lat big = -36

Stopped. Moved lat big to -56 to realign
720225 beam
Angular scan -1000 urad / 1000 urad starting at -1200 urad, biglat = -36mm
720227 beam ANGULAR SCAN



x-axis = rot position
y-axis = deflection

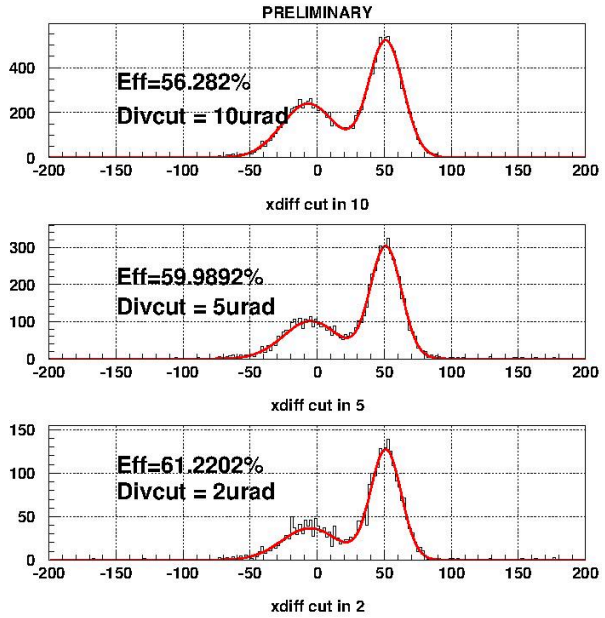
720228 latbig = -36, rot = -1133 CHANNELING high stat 432 kevents
720229 pede
720230 latbig = -36, rot = -1133 CHANNELING high stat 511k events

24/08/2023

Modified DAQ to increase nr of triggers per spill → taken out all the CAEN stuff (apart from the controller). We go from 3000 to almost 6000. This is the maximum with 4 VRBs (the dead time is mainly due to the init of the boards at each event).
The percentage of single clusters is between 25% and 30% depending on the intensity

720232 latbig = -36, rot = -1133 CHANNELING

Summing 720228+720230+720232 →



x axis = deflection

720233 latbig = -36, rot = -1138 CHANNELING
720234 latbig = -36, rot = -1128 CHANNELING
720235 latbig = -36, rot = -1128 CHANNELING

Connected pollux with vertical motor to scan the vertical position
720239 pede
720243 vertical scan -10/-100mm in steps of 10mm 10kev/step (numbers have to be divided by 2 because of the motor encoder, to find the real numbers)
Waiting for beam …..
Vertical position of the holder found @vertical = -50
720246 finer scan in steps of 2mm from -60 to -40 → holder edge found at -41

720247 rot=-1133, z=-6mm, lat=-36mm
The beam has moved → the CCC has made tests during the afternoon
720248 rot=-1128, z=-6mm, lat=-36mm
We miss one spill → reduced the maximum nr of events per spill to 4000.
720249 rot=-1128, z=-6mm, lat=-36mm
Wrong way, let's move on the other side
720250 rot=-1138, z=-6mm, lat=-36mm → difficult

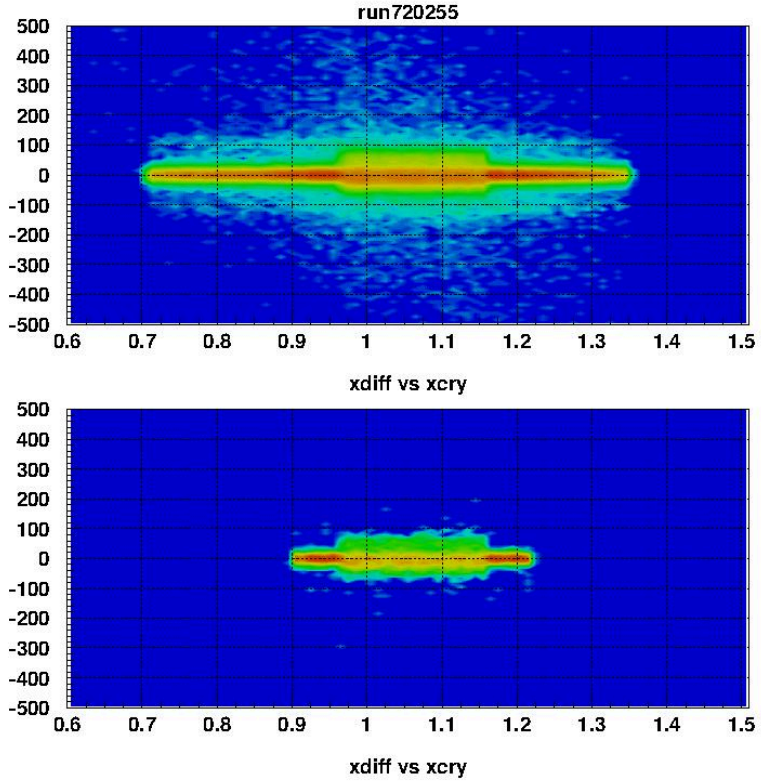
Let's take an amorphous run moving lat to +10mm (Absolute)
720251 amorphous to ALIGN

720252 rot = -1138 → wrong way
720253 rot = -1133 → wrong way

720254 rot = -1143 → not enough
720255 rot = -1153 long run – usable CHANNELING

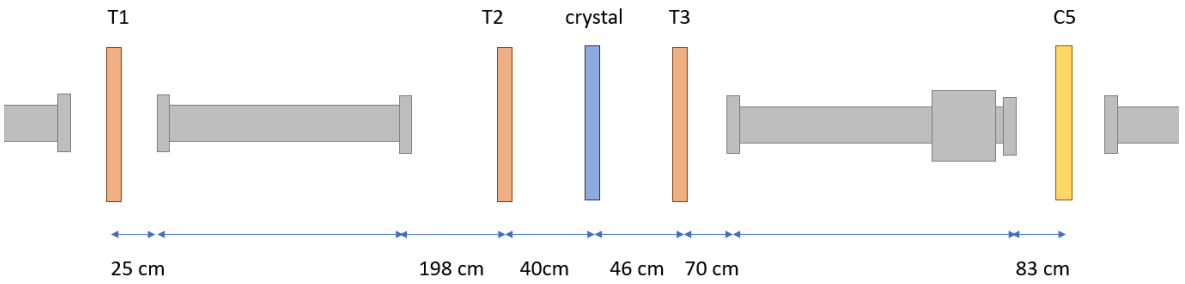
Refining alignment + narrowing the trigger region →

720056 rot = -1150, trig=142-162 on VA2TA instead of 122-262
720057 rot = -1150, trig=152-252 on VA2TA
720058 rot = -1150, trig=162-242 on VA2TA
700259 pede
720060 rot = -1150, trig=158-240 on VA2TA CHANNELING
720061 rot = -1150, trig=168-230 on VA2TA CHANNELING → >1 Mevents



25/08/2023

720262 pede before access
720263/4/5 pede after access



from 720267-720274 trials with beam

720274 lat = -27 we see the crystal
720275 lat = -25 wrong direction → need to enter because we are at the limit with the lat motor

Intervention to:

- move BC5 after around 5m
- set APC thresh to 50 mV to use as a veto → its signal is inverted and stretched to 10us

720297 pede
720298 crystal on beam border lat = -17
720299 crystal in center lat = -19
720300 lat = -10, amorphous
720301 lat = -10, amorphous

720302 crystal is center
720304 vertical scan 0 / -14mm in steps of -2 mm (remember the factor 2 because of the encoder)

720305 vertical scan 0 / -10mm in steps of -1 mm (remember the factor 2 because of the encoder). Start at -18 → found holder at -23 → crystal center at -14 mm (factor 2!)

Crystal center → lat = -19, biglat (vertical) = -14

Single DAQs to see where we are (volume reflection?) and when we exit from this condition

720307 rot = -750 → in volume reflection
720308 rot = -1750
720309 rot = -11750
720310 rot = -6750
720311rot = -4250
720312/15 rot = -5500
720313/14 rot = -4875
720316 rot = -6125

720317 rot = -4865
720318 rot = -4885
720319 rot = -4875
720320 rot = 3127

720321 → 720325 crap

720326 senza cristallo
720327 con cristallo, rot = 5000
720328 rot = -750 (volume reflection)
720329 rot = +250
720330 rot = 8250 (forse siamo in volume reflection)
720331 rot = -11750 (secondo noi vero amorfo)
720332 rot = - 6250 (amorfo)
720333 rot = -3500

Ci siamo accorti che con il cristallo abbiamo una deflessione anche in y. Magari perché siamo vicini all'holder. Proviamo a verificare la posizione in y

720334 crap
720335 scan lat big (verticale) da 0 a -40, con step = 2mm, rot = -750

720338 pede post access

720339

Scan across the dinner
720340 back to -4200 (si parte da -3200) to avoid channelling
step 250
10k evts per step

Since we tale 300/4000 good events, we desire 10.000*4.000/300 ~ 150.000 events to acquire
Should be 12' per step

720341 rot = -700, cradle = -1500, singola
720342 rot = -800, cralde = -1500, scan

720344 rot = -950, single daq

720345 da -1500 ins, sempre 150.000 evts, step 50. Stessa strategia usata stasera a cena.
Craddle fissa a -1500
Aborted soon

720346 pede

Scan across the night
720347
Same scan, but for real. rot scan, from -1500 to positive, 150.000 evts, 50urad per step.
Cradle at -1500

720348 pede
720349 crad = -3500, scan rot from -900 to positive 150kev/step 50urad step

Craddle -10.000

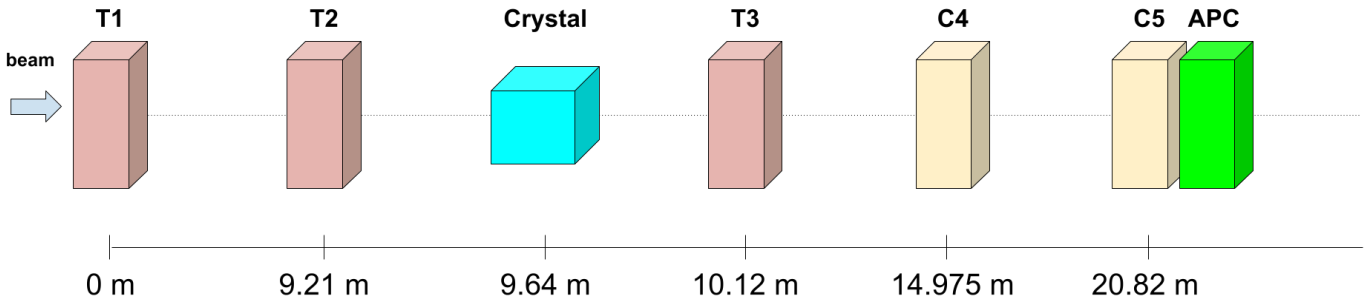
720359 crap
720360 trigger semplice per vedere i single cluster (della quarta camera leggiamo prima la x e poi la y) (20% single cluster, vorremmo scendere in intensità)
720361 con anche il veto (APC) (25% single cluster)



ASCII FILES STRUCTURE MODIFICATION TO INCLUDE 3 SILICON TELESCOPES + 2 SILICON BEAM CHAMBERS from run 720360

Ascii structure follows (as soon as will be ready)
THE STRUCTURE IS THE SAME
SIMPLY THE 4TH DETECTOR IS BC4 (THE CLOSEST ONE) AND NOT ANYMORE BC5

- THE ASCII STRUCTURE IS THE SAME
- THE DISTANCE OF THE LAST CHAMBER CHANGED



720362 **removed crystal to align beam chambers (lat = +10 absolute)**
720363 trigger piccolo, portato dentro il cristallo (lat = -19)

Sulla camera centrale qualcosa si vede → andava a sbattere contro la pipe dietro

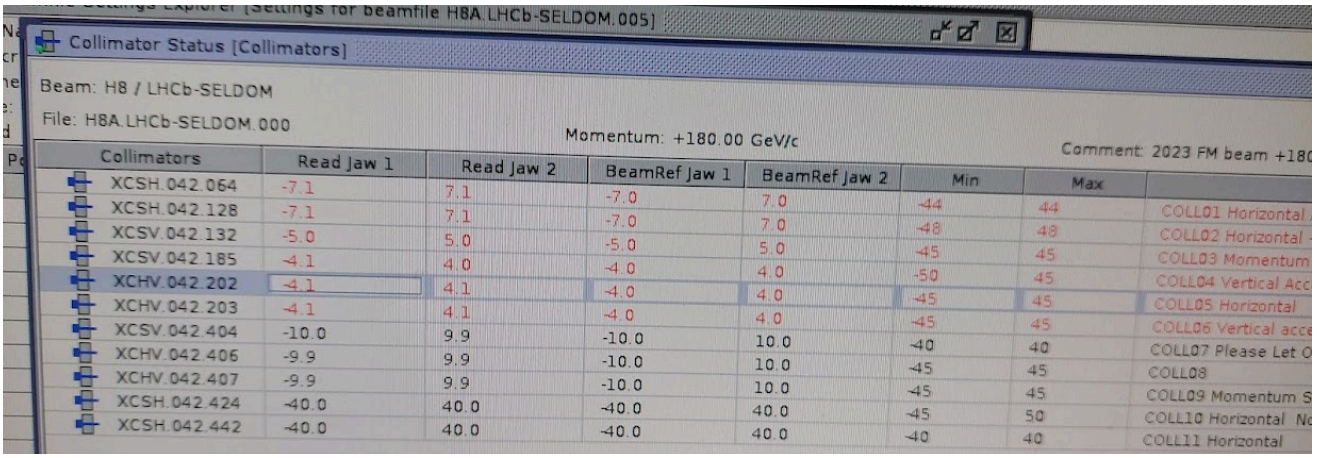
720364 scan. rot da -200 a 10000, step 50, nr eventi per step = 150000 (start = -600) crad = -10000

Scarsi ha giocato un po' col fascio. Siamo passati da 670K a 530K a 390K. Abbiamo anche tolto il taglio messo in Galore per il telescopio che derivava.
Single cluster 50%

720365 pede
720366
720367 **scan rot da 0 a 10000, start -10000, step 50, nr eventi 150000**

Called Piquet (67500, Bastien Rae) for asking a lower intensity beam. Told us to load a file with a higher intensity
Called CCC (77500) to ask suggestions about which collimator to use. They told that know exactly what to do but couldn't tell us

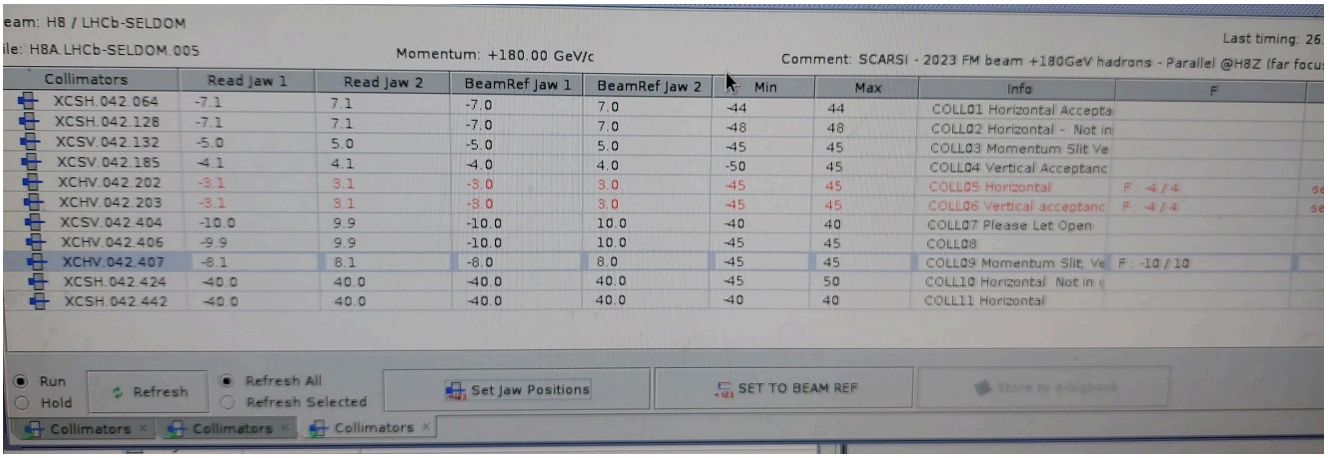
Starting point, file 000



Modifiche a sentimento che hanno permesso di raggiungere ~ 50% of good triggers. Momentum slit was suggested by bastien but apparently helps little.
Collimators 202 and 203 closed of +-1mm did the magic

720369 lost control of the cradle, run in rot = -650, position of the cradle not clear

720370 pede



We have one hour before lhcb velo intervention.
Crad at -20.000. Cannot reach -20.000

720367 rot = -650 crad=-10000

720370 pede
720371 rot da -250 (che corrisponde a -900) a 1000, step 50 urad, 100000 per step → detector moved during intervention on goniometer

720372-720373 crap

720374 run di allineamento

720375 0 era sul bordo. Parto da 0, vado in assoluto +1mm di lat, per verificare che siamo centrati. Infatti non lo siamo
720376 -2 relativo (=-1 abs)
720377 Allarghiamo il trigger
720378 Stringiamo a sinistra il trigger (142-282 sul va2ta)
720379 Stringiamo ancora 162-282
720380/720381 ?? move crystal. abs +10 - trigsel 2
720382 106-236 sul primo [Intervallo di strip in x su cui triggerare](10 strip verso sx, prima era 116-246)
720383 back trig sel 8, try to fix va2ta strip. start with 100-300
720384 trigger → 80-280
720385 trigger → 60-240
720386 trigger →100-280
720387 trigger → 200-300

720388 Riportiamo dentro il cristallo
720389 lat = +0.4 per centrare il cristallo
720390 load_reg_chan_VA2TA 190 310
720391 cristallo OK
720392 pede. Tele 1 prendeva un sacco di 2-cluster. Con un pede abbiamo risolto
720393 scan partendo da ang = -250, da 0 a 1000, 10000 eventi per step

720394 scan da -500, da 0 a 1000, 10000 eventi per step

La culla dà nuovamente problemi. Abbiamo provato a risistemare la guida ottica
La craddle non rispondeva. Decidiamo di non controllarla da remoto ma solo in presenza. Rispetto al preallineamento, usando la livella, ci siamo inclinati di ~3.2° (Ricordo io. Andrea a voce disse ~4°)

720395-720397 pede

lat +10 per togliere il cristallo
720399 trigger stretto come prima (funziona :))
720400 rimettiamo il cristallo dentro (lat = 0) → posizione cristallo 0.94, -1.13
720401 scan rot da -1000 a 1000 step 100, 50k eventi per step
Channelling appared at step 16. Alla faccia di tutti i miscredenti!
No skew plane presents

720402 Ang abs 3650 single daq, trig Sili*!APC. Aborted
720403 Pede
720404 Ang abs 3650 single daq, trig Sili*!APC. LONG RUN ACCROSS NIGHT
720405 Pede
720406 Same scan, but restarted because of high intensity beam change

720407 pede

720408 allarghiamo il trigger → 170,330

Divout-divin
trigger largo
Eventi tra 3500 e 4000 0.14%
Eventi tra 4000 e 4500 0.11%
Eventi tra 5500 e 6000 0.19%
trigger stretto
Eventi tra 3500 e 4000 0.12%
Eventi tra 4000 e 4500 0.20%
Eventi tra 5500 e 6000 0.26%

720409 same trigger. Muoviamo in y il cristallo a -18 assoluto

MOVED BC4 2cm TOWARD SALEVE. THIS WAY, THE BEAM CENTER SHOULD BE AT ~ 3cm
720410 pede
720411 Moving crystal +10 abs in lat (starting from 0)
RUN DI ALLINEAMENTO - ALIGNMENT RUN

FROM NOW ON, A RE-ALIGNMENT OF THE BC4 HAS TO BE PERFORMED

720412 Crystal again on beam. Lat = 0 abs

720413 pede
720414 rot = 7850 (zona del buco) per vedere se vedevamo qualcosa di strano

720415 Scan rot nella zona ~~del buco~~ della volume capture. Partiamo da 7650, 0 a 2000 a passi di 50urad, 1 milione di eventi per step (run per la notte)

2023-08-28
LHCB-VELO team gifted us 30 mins of low intensity. Actually in the first 7 mins, collimators 202 and 203 were +-4mm, then +-3mm as two days ago

CRYSTAL 2C-4
(anodic bonding)

720416 pede
720417 scan y nella zona del channeling. Partiamo da -8, andiamo da 0 a -40 a passi di -4, 20k eventi per step. Ricordiamoci il fattore due!

720418 iniziato scan più fine in y → abortito

It seems this night we moved in the wrong direction. Now we go at y=-8 (with the usual factor 2) and use the last 10 mins of low intensity to acquire statistic
720419 Single daq, y = -8

9.50: going inside for crystal change
Motor position before removing

720420-720421 pede
720422 single daq, rot = 2750, verticale (lat big) = -36, lat = 0
720423
720424 scan su y (ricordiamo il fattore 2), partenza -36, da 0 a -24, step = -4, 10k eventi per step, rot = 2750
720425 trigger a -290
720426 trigger a -310
720427 lat = -3mm. trigger a -350
720428 ??
720429 abilita tutto il va2ta
720430 lat = 6
720431 lat = 1
720432 lat = 0.5 → inizio cristallo 0.91
720433 lat = 0 → inizio cristallo 0.85
720434 lat = 0.5, stringiamo il trigger 150 da una parte
720435 stringiamo a 200 (stavamo tagliando da due parti)
720436 partiamo dal cip di mezzo → 128-256. Abbiamo tagliato il cristallo dalla parte del vetro
720437 128-220 abbiamo stretto troppo
720438 128-240 Da questa parte va bene
720439 140-240 Dobbiamo tagliare di più
720440 160-240
720441 180-240
720442 190-240
720443 190-250 trigsel1 non siamo perfettamente allineati…
trigger sel 1 VA2TA → 190-250
load_reg_chan_VA2TA 190 250 1 384 1 0 0

trig2023_select 1 1

720444 lat = 10 abs run di allineamento
720445 lat = 0.5. scan rot partenza = +2750, da 0 a 2000, passo 50urad, 48k per step

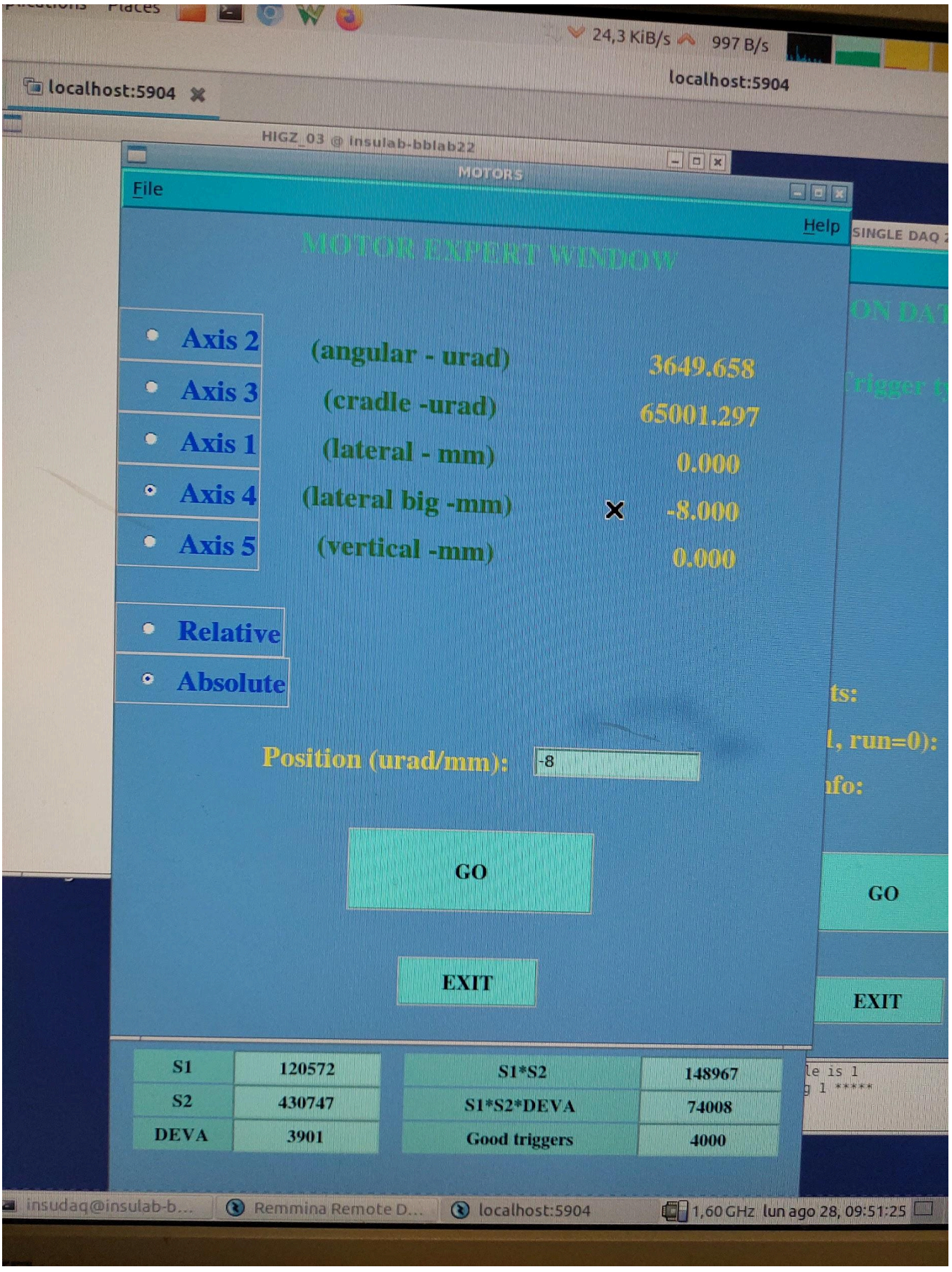
12 spills per step are needed. We took 2 spills every minute ⇒6 min per step. 40 step * 6 min = 240 min = 4 hours

720446 pede
720447 run di allineamento lat = 10
720448 rot abs 2800 and a quick single daq

720449 gonio rot 2800→2760 (-40 rel)
720450 Wrong direction. Moving +80 i.e. +2840 abs **IT IS A REAL MASTERPIECE**
720451 same, without apc — this has solved

Installed first crystal

720452 pede
720453 pede


24,3 KiB/s 997 B/s
localhost:5904
localhost:5904
HIGZ_03 @ insulab-bblab22
MOTORS
File
Help
MOTOR EXPERT WINDOW
Axis 2
Axis 3
Axis 1
Axis 4
Axis 5
(angular - urad)
(cradle -urad)
(lateral - mm)
(lateral big -mm)
(vertical -mm)
3649.658
65001.297
0.000
-8.000
0.000
Relative
Absolute
Position (urad/mm):
-8
GO
EXIT
S1
120572
S2
430747
DEVA
3901
S1*S2
148967
S1*S2*DEVA
74008
Good triggers
4000
SINGLE DAQ
GO
EXIT
insudaq@insulab-b...
Remmina Remote D...
localhost:5904
1,60 GHz lun ago 28, 09:51:25

Crystal realized with anodic bonding has torsion of about ZERO urad/mm
Crystal mechanically bent has torsion of about 40 urad/mm, see below plots.

For 2C-4 crystal (anodic bonding) in the data analysis, the center of angular cut must be set at -9 urad (run 720451).

CRYSTAL 2C-3

29/08/23 00:10
Crystal re-installed by Andrea and Pascal. After laser alignment the crystal was already in channeling orientation!

720454	lat = 20 to move out the crystal → alignment
720455	lat = -8. Crystal back in. Single daq
720456	lat = -2 abs.
720457	lat = -0.5 abs. Crystal centered!
720458	scan rot partendo da 10541, da 0 a 1000, step 50, 48k/step (we didn't moved back of 7mrad)
720459	scan rot partendo da 3590, da 0 a 100, step 50, 48k/step → **already in channeling!**
720460	single daq. lat = -50 rel, 3590 abs. (and we are not in channeling :))
720461	Miniscan. lat = +50 rel, da 0 a 1000, step 50, 48k/step
720462	rot = 3790. High statistics CHANNELING
720463	pede

Runs to study VR

720464	scan rot to enter in volume reflection: from 3790 + 1000 mrad in steps of 75ura, 40kev per step (20k good)
720465	scan rot: from 3790 -215 mrad in steps of -75 urad, 40k per step
720466	scan rot: from 103000 +1000 mrad in steps of 75 urad, 40k per step
720467	single daq. rot = 7500 → center of VR
720468	single daq. rot = 1000 → amorphous
720469	pede
720470	rot = 3790. High statistics CHANNELING

720471	pede
720472	rot=1000 high stat amorphous